

Mechanics 2

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2.4.1 Linear momentum

Momentum and impulse

For an object of mass m moving at velocity $\vec{v}$, its momentum is defined as

$$\vec{p} = m\vec{v}.$$

The concept of momentum is derived as a restatement of Newton's Second Law.

$$\vec{F} = m\vec{a} = m \frac{d\vec{v}}{dt} = \frac{d}{dt} \underbrace{(m\vec{v})}_{=\vec{p}}.$$

In other words, force is equal to the **rate of change of momentum**.

$$\int_{\vec{p}_1}^{\vec{p}_2} d\vec{p} = \int_{t_1}^{t_2} \vec{F} dt$$
$$\Delta\vec{p} = \vec{p}_2 - \vec{p}_1 = \int_{t_1}^{t_2} \vec{F} dt.$$

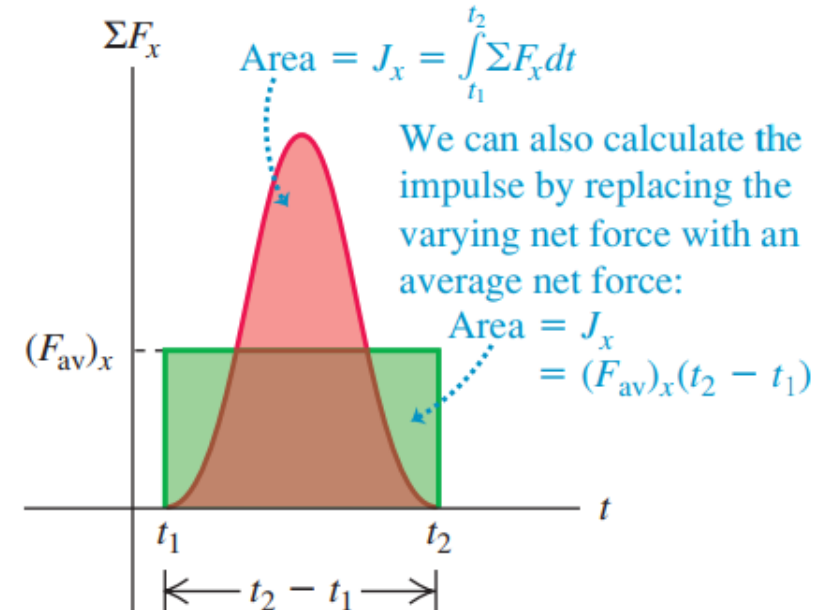
We define **impulse** as the change in momentum

$$\vec{J} = \Delta\vec{p} = \int_{t_1}^{t_2} \vec{F} dt.$$

8.3 The meaning of the area under a graph of ΣF_x versus t .

(a)

The area under the curve of net force versus time equals the impulse of the net force:



2.4.1 Linear momentum

Additionally, kinetic energy K can be related to momentum as follows:

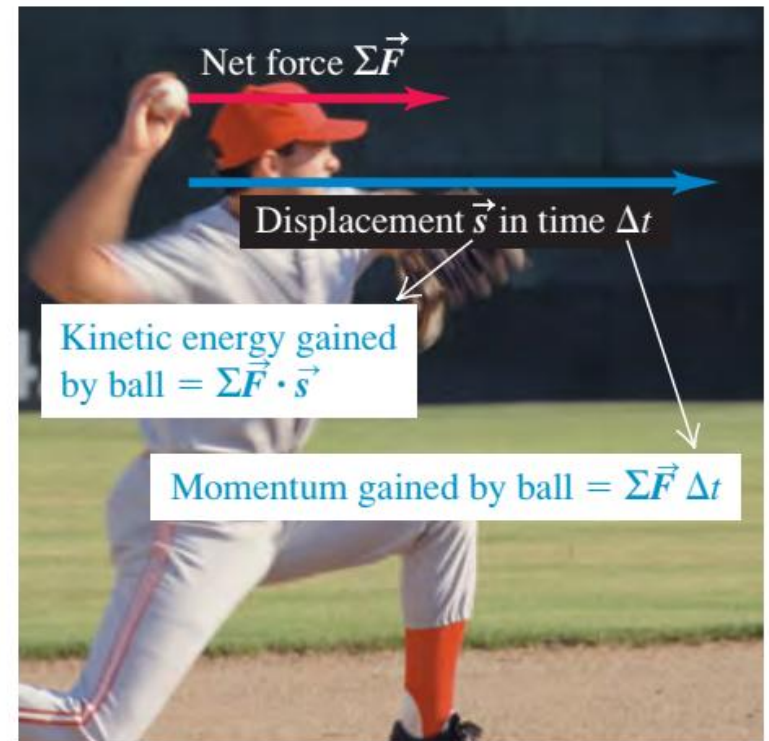
$$\begin{aligned} K &= \frac{1}{2}mv^2 = \frac{1}{2}m\vec{v} \cdot \vec{v} = \frac{1}{2m}m^2\vec{v} \cdot \vec{v} = \frac{1}{2m} (m\vec{v}) \cdot (m\vec{v}) \\ &= \frac{1}{2m}\vec{p} \cdot \vec{p} \\ &= \frac{1}{2m}p^2. \end{aligned}$$

So, the kinetic energy is related to momentum by

$$K = \frac{p^2}{2m},$$

where $p = \sqrt{\vec{p} \cdot \vec{p}}$ is the magnitude of $\vec{p}$.

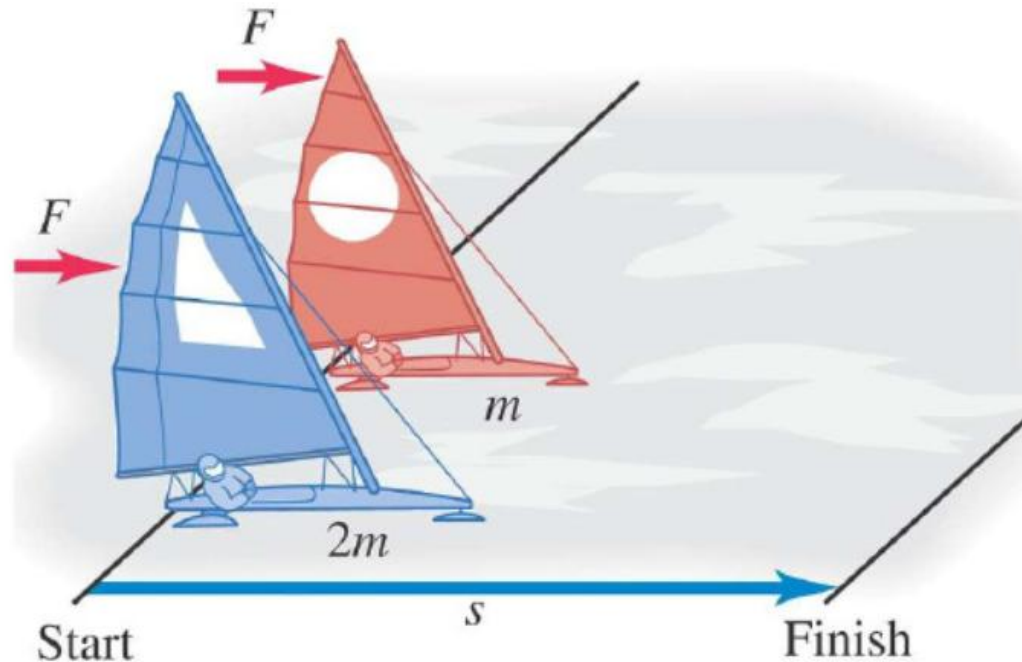
8.4 The *kinetic energy* of a pitched baseball is equal to the work the pitcher does on it (force multiplied by the distance the ball moves during the throw). The *momentum* of the ball is equal to the impulse the pitcher imparts to it (force multiplied by the time it took to bring the ball up to speed).



2.4.1 Linear momentum

Example 4.1.1:

Two ice-boats hold a race on a frictionless frozen lake. Boat 1 has mass m and boat 2 has mass $2m$, with identical sails. Hence the wind exerts the **same constant horizontal force $\vec{F}$ on each boat**. They start from rest and cross the finish line a distance s away. Which boat crosses the finish line with greater (a) kinetic energy? (b) momentum?



2.4.1 Linear momentum

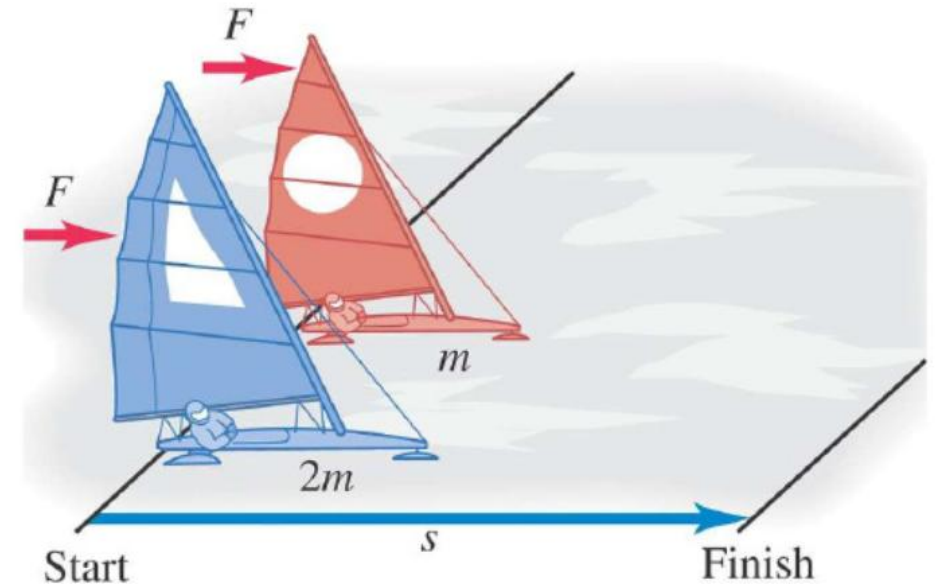
According to the work-energy theorem,

$$W = Fs = K_f - K_i = K_f,$$

*since both start from rest, $K_i = 0$. Since F and s are the same for both boats, therefore K_f are also the **same for both boats**. Since both boats have the same KE, $K_{1f} = K_{2f}$, and using $K = \frac{p^2}{2m}$,*

$$\begin{aligned} K_{1f} &= K_{2f} \\ \frac{p_1^2}{2m} &= \frac{p_2^2}{2(2m)} \quad \rightarrow \quad p_2^2 = 2p_1^2 \end{aligned}$$

Therefore Boat 2 has larger momentum than Boat 1.

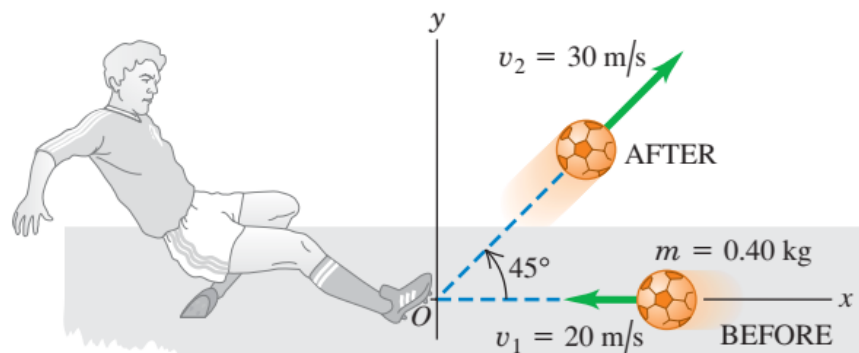


2.4.1 Linear momentum

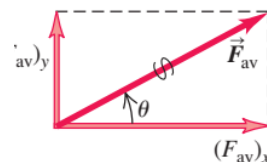
Example 8.3 Kicking a soccer ball

A soccer ball has a mass of 0.40 kg. Initially it is moving to the left at 20 m/s, but then it is kicked. After the kick it is moving at 45° upward and to the right with speed 30 m/s (Fig. 8.7a). Find the impulse of the net force and the average net force, assuming a collision time $\Delta t = 0.010$ s.

(a) Before-and-after diagram



(b) Average force on the ball



SOLUTION

IDENTIFY and SET UP: The ball moves in two dimensions, so we must treat momentum and impulse as vector quantities. We take the x -axis to be horizontally to the right and the y -axis to be vertically upward. Our target variables are the components of the net impulse on the ball, J_x and J_y , and the components of the average net force on the ball, $(F_{av})_x$ and $(F_{av})_y$. We'll find them using the impulse–momentum theorem in its component form, Eqs. (8.9).

EXECUTE: Using $\cos 45^\circ = \sin 45^\circ = 0.707$, we find the ball's velocity components before and after the kick:

$$v_{1x} = -20 \text{ m/s} \quad v_{1y} = 0$$

$$v_{2x} = v_{2y} = (30 \text{ m/s})(0.707) = 21.2 \text{ m/s}$$

From Eqs. (8.9), the impulse components are

$$\begin{aligned} J_x &= p_{2x} - p_{1x} = m(v_{2x} - v_{1x}) \\ &= (0.40 \text{ kg})[21.2 \text{ m/s} - (-20 \text{ m/s})] = 16.5 \text{ kg} \cdot \text{m/s} \end{aligned}$$

$$\begin{aligned} J_y &= p_{2y} - p_{1y} = m(v_{2y} - v_{1y}) \\ &= (0.40 \text{ kg})(21.2 \text{ m/s} - 0) = 8.5 \text{ kg} \cdot \text{m/s} \end{aligned}$$

From Eq. (8.8), the average net force components are

$$(F_{av})_x = \frac{J_x}{\Delta t} = 1650 \text{ N} \quad (F_{av})_y = \frac{J_y}{\Delta t} = 850 \text{ N}$$

The magnitude and direction of the average net force $\vec{F}_{av}$ are

$$F_{av} = \sqrt{(1650 \text{ N})^2 + (850 \text{ N})^2} = 1.9 \times 10^3 \text{ N}$$

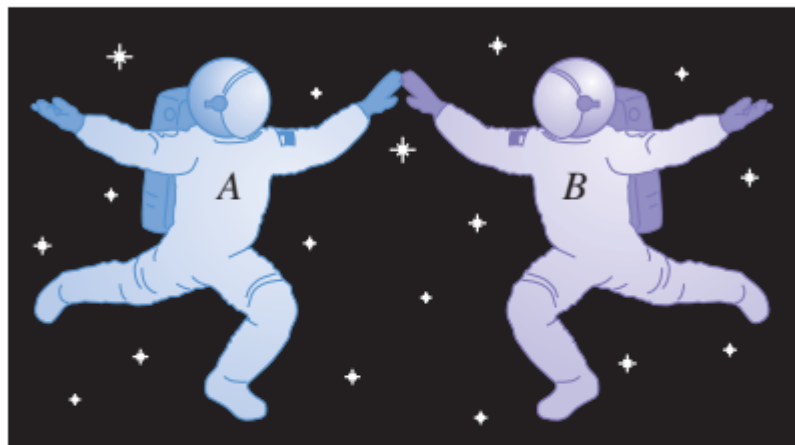
$$\theta = \arctan \frac{850 \text{ N}}{1650 \text{ N}} = 27^\circ$$

The ball was not initially at rest, so its final velocity does *not* have the same direction as the average force that acted on it.

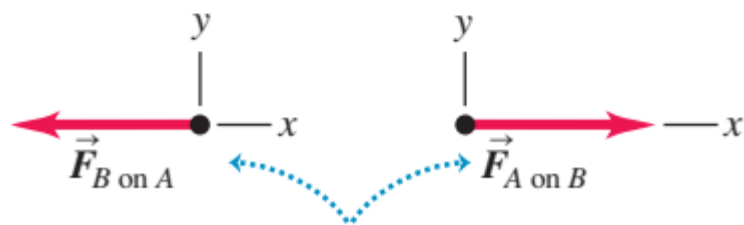
EVALUATE: $\vec{F}_{av}$ includes the force of gravity, which is very small; the weight of the ball is only 3.9 N. As in Example 8.2, the average force acting during the collision is exerted almost entirely by the object that the ball hit (in this case, the soccer player's foot).

2.4.2 Conservation of momentum

8.8 Two astronauts push each other as they float freely in the zero-gravity environment of space.



No external forces act on the two-astronaut system, so its total momentum is conserved.

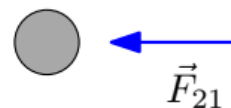


The forces the astronauts exert on each other form an action–reaction pair.

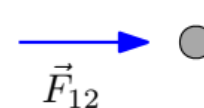


An instantaneous collision between two particles.

Particle 1 FBD:



Particle 2 FBD:



The only forces are those forces where 1 and 2 interact with each other, $\vec{F}_{21}$ and $\vec{F}_{12}$. We assume there are **no external forces**, $\vec{F}_{\text{ex}} = 0$.

From the FBD of particle 1: $\vec{F}_{21} = \frac{d\vec{p}_1}{dt}$.

From the FBD of particle 2: $\vec{F}_{12} = \frac{d\vec{p}_2}{dt}$.

Adding up the above two equations, $\frac{d\vec{p}_1}{dt} + \frac{d\vec{p}_2}{dt} = \vec{F}_{21} + \vec{F}_{12}$

$$\frac{d}{dt} (\vec{p}_1 + \vec{p}_2) = \vec{F}_{21} + \vec{F}_{12}.$$

By Newton's 3rd Law, $\vec{F}_{21} = -\vec{F}_{12}$.

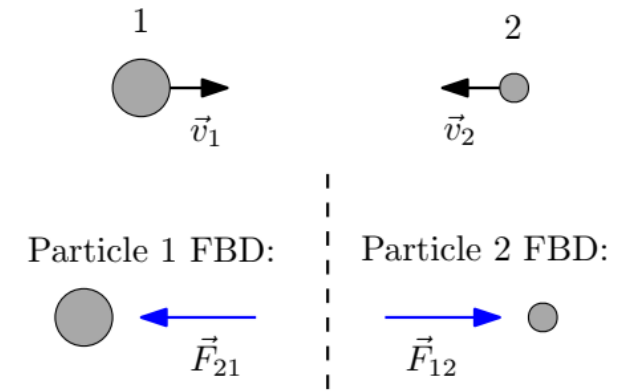
$$\frac{d}{dt} \underbrace{(\vec{p}_1 + \vec{p}_2)}_{=\vec{p}_{\text{tot}}} = 0$$

$$\boxed{\frac{d\vec{p}_{\text{tot}}}{dt} = 0}$$

2.4.2 Conservation of momentum

$$\frac{d}{dt} \underbrace{(\vec{p}_1 + \vec{p}_2)}_{=\vec{p}_{\text{tot}}} = 0$$

If the vector sum of **the external forces on a system** is **zero**, the total momentum of the system is **constant**



Since total momentum is constant, the value before and after collision should be the same.

$$\begin{aligned} (\vec{p}_1 + \vec{p}_2 \text{ before}) &= (\vec{p}_1 + \vec{p}_2 \text{ after}) \\ m_1 \vec{u}_1 + m_2 \vec{u}_2 &= m_1 \vec{v}_1 + m_2 \vec{v}_2. \end{aligned}$$

2.4.2 Conservation of momentum

Problem-Solving Strategy 8.1 Conservation of Momentum



IDENTIFY *the relevant concepts:* Confirm that the vector sum of the external forces acting on the system of particles is zero. If it isn't zero, you can't use conservation of momentum.

SET UP *the problem* using the following steps:

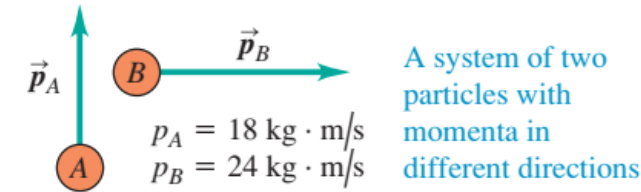
1. Treat each body as a particle. Draw “before” and “after” sketches, including velocity vectors. Assign algebraic symbols to each magnitude, angle, and component. Use letters to label each particle and subscripts 1 and 2 for “before” and “after” quantities. Include any given values such as magnitudes, angles, or components.
2. Define a coordinate system and show it in your sketches; define the positive direction for each axis.
3. Identify the target variables.

EXECUTE *the solution:*

1. Write an equation in symbols equating the total initial and final x -components of momentum, using $p_x = mv_x$ for each particle. Write a corresponding equation for the y -components. Velocity components can be positive or negative, so be careful with signs!
2. In some problems, energy considerations (discussed in Section 8.4) give additional equations relating the velocities.
3. Solve your equations to find the target variables.

EVALUATE *your answer:* Does your answer make physical sense? If your target variable is a certain body's momentum, check that the direction of the momentum is reasonable.

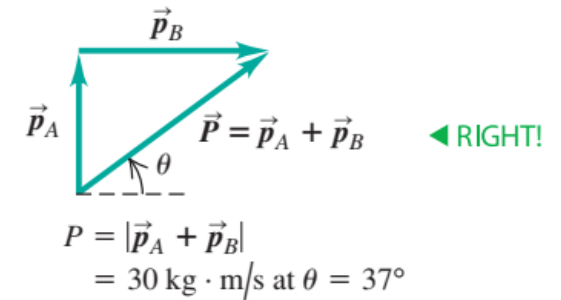
8.10 When applying conservation of momentum, remember that momentum is a vector quantity!



You **CANNOT** find the magnitude of the total momentum by adding the magnitudes of the individual momenta!

$$P = p_A + p_B = 42 \text{ kg} \cdot \text{m/s} \quad \text{◀ WRONG}$$

Instead, use vector addition:



2.4.2 Conservation of momentum

Example 4.2.1:

Two carts of different masses move toward each other on a friction-less surface. Cart 1 has mass $m_1 = 0.50$ kg and cart 2 has mass $m_2 = 0.30$ kg. After they collide, mass 2 has a final velocity of $+2.0 \text{ m s}^{-1}$. What is the final velocity of mass 1?

Before collision

$$u_1 = 2.0 \text{ m s}^{-1}$$

$$u_2 = -2.0 \text{ m s}^{-1}$$



After collision

$$v_1 = ?$$

$$v_2 = 2.0 \text{ m s}^{-1}$$



2.4.3 Collision- elastic and inelastic collision

Collisions in one dimension

- **Elastic collisions:** Collisions where kinetic energy is conserved.
- **Inelastic collisions:** Collisions where kinetic energy is **not** conserved. If the two objects stick together after collision, it is a **completely inelastic collision**.

Elastic collisions

Since this is a collision, linear momentum is conserved,

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2.$$

$$m_1 (u_1 - v_1) = m_2 (v_2 - u_2) .$$

$$u_1 + v_1 = v_2 + u_2$$

$$(u_1 - u_2) = -(v_1 - v_2)$$

And for elastic collisions, kinetic energy is also conserved,

$$\frac{1}{2}m_1 u_1^2 + \frac{1}{2}m_2 u_2^2 = \frac{1}{2}m_1 v_1^2 + \frac{1}{2}m_2 v_2^2.$$

$$m_1 (u_1^2 - v_1^2) = m_2 (v_2^2 - u_2^2)$$

$$m_1 (u_1 - v_1) (u_1 + v_1) = m_2 (v_2 - u_2) (v_2 + u_2) .$$

This formula is only true for elastic collisions!

2.4.3 Collision- elastic and inelastic collision

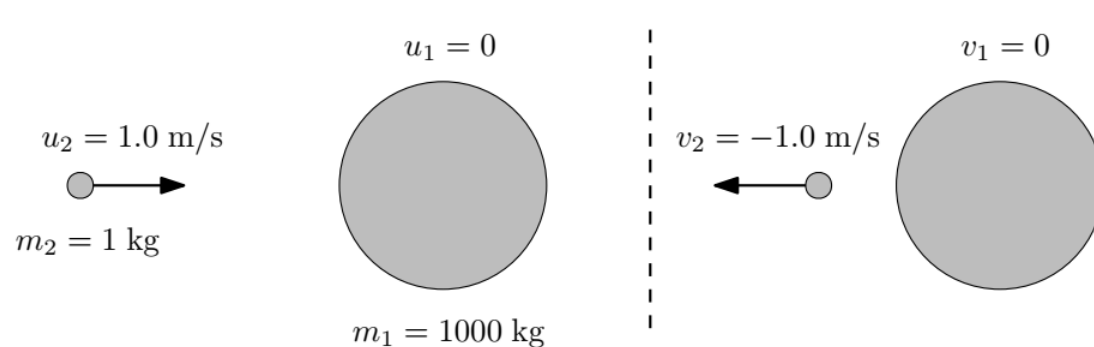
Collisions in one dimension

Elastic collisions

$$m_1 (u_1 - v_1) = m_2 (v_2 - u_2). \quad v_2 = u_1 - u_2 + v_1, \quad \text{or} \quad v_1 = v_2 + u_2 - u_1.$$

$$v_1 = \frac{2m_2 u_2 + (m_1 - m_2)u_1}{m_1 + m_2}, \quad \text{or} \quad v_2 = \frac{2m_1 u_1 - (m_1 - m_2)u_2}{m_1 + m_2}.$$

When a small mass collides with a heavy object at rest



$$v_1 = \frac{2m_2 u_2}{m_1 + m_2} \quad v_2 = \frac{-(m_1 - m_2)u_2}{m_1 + m_2}.$$

$$m_2 \ll m_1$$

$$v_1 \simeq 0, \quad v_2 \simeq -u_2.$$

2.4.3 Collision- elastic and inelastic collision

Collisions in one dimension

Elastic collisions

Example 8.4 Recoil of a rifle

A marksman holds a rifle of mass $m_R = 3.00$ kg loosely, so it can recoil freely. He fires a bullet of mass $m_B = 5.00$ g horizontally with a velocity relative to the ground of $v_{Bx} = 300$ m/s. What is the recoil velocity v_{Rx} of the rifle? What are the final momentum and kinetic energy of the bullet and rifle?

Example 4.3.1:

Two gliders with masses $m_A = 0.50$ kg and $m_B = 0.30$ kg with ideal spring bumpers collide elastically. What are the final velocities of the two gliders?

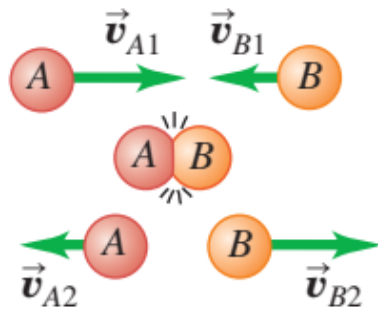
2.4.3 Collision- elastic and inelastic collision

Collisions in one dimension

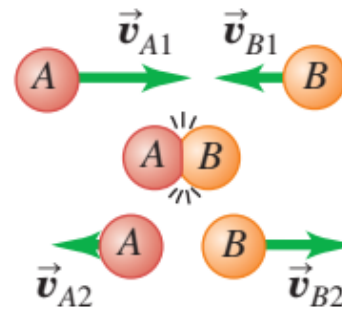
Inelastic collisions, completely inelastic collision

8.20 Collisions are classified according to energy considerations.

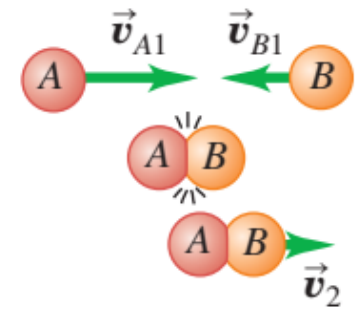
Elastic:
Kinetic energy
conserved.



Inelastic:
Some kinetic
energy lost.



Completely inelastic:
Bodies have same
final velocity.



2.4.3 Collision- elastic and inelastic collision

Collisions in two dimension

Example 4.4.1:

Fig. 4.6 shows an elastic collision of two billiard balls on a frictionless surface. Ball A has an initial velocity of 4.00 m/s in the positive x -direction and a final velocity of 2.00 m/s in an unknown direction α . Ball B is initially at rest. Find the final speed v_B of B and the angles α and β .

2.4.3 Collision- elastic and inelastic collision

Collisions in two dimension

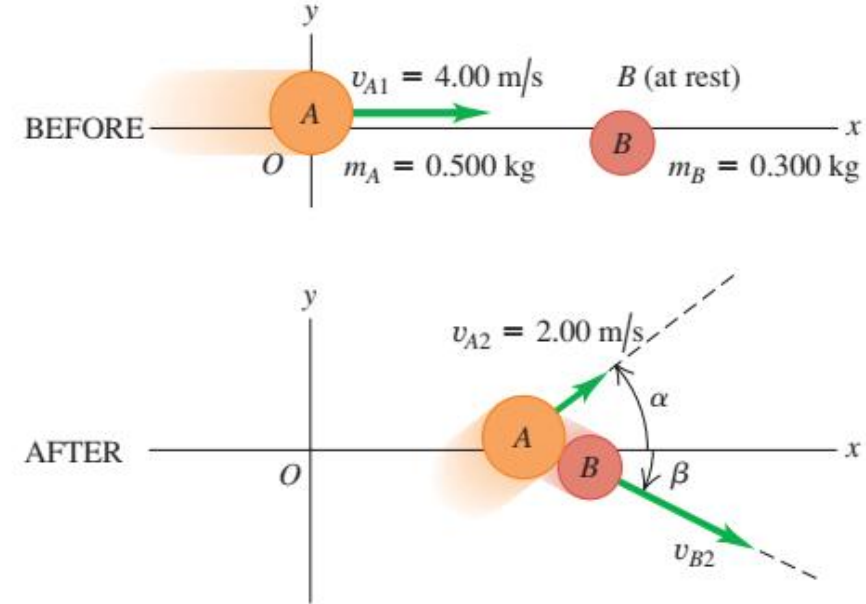
IDENTIFY and SET UP: We'll use the equations for conservation of energy and conservation of x - and y -momentum. These three equations should be enough to solve for the three target variables given in the problem statement.

EXECUTE: The collision is elastic, so the initial and final kinetic energies of the system are equal:

$$\begin{aligned}\frac{1}{2}m_A v_{A1}^2 &= \frac{1}{2}m_A v_{A2}^2 + \frac{1}{2}m_B v_{B2}^2 \\ v_{B2}^2 &= \frac{m_A v_{A1}^2 - m_A v_{A2}^2}{m_B} \\ &= \frac{(0.500 \text{ kg})(4.00 \text{ m/s})^2 - (0.500 \text{ kg})(2.00 \text{ m/s})^2}{0.300 \text{ kg}} \\ v_{B2} &= 4.47 \text{ m/s}\end{aligned}$$

Conservation of the x - and y -components of total momentum gives

$$\begin{aligned}m_A v_{A1x} &= m_A v_{A2x} + m_B v_{B2x} \\ (0.500 \text{ kg})(4.00 \text{ m/s}) &= (0.500 \text{ kg})(2.00 \text{ m/s})(\cos \alpha) \\ &\quad + (0.300 \text{ kg})(4.47 \text{ m/s})(\cos \beta) \\ 0 &= m_A v_{A2y} + m_B v_{B2y} \\ 0 &= (0.500 \text{ kg})(2.00 \text{ m/s})(\sin \alpha) \\ &\quad - (0.300 \text{ kg})(4.47 \text{ m/s})(\sin \beta)\end{aligned}$$



These are two simultaneous equations for α and β . We'll leave it to you to supply the details of the solution. (*Hint:* Solve the first equation for $\cos \beta$ and the second for $\sin \beta$; square each equation and add. Since $\sin^2 \beta + \cos^2 \beta = 1$, this eliminates β and leaves an equation that you can solve for $\cos \alpha$ and hence for α . Substitute this value into either of the two equations and solve for β .) The results are

$$\alpha = 36.9^\circ \quad \beta = 26.6^\circ$$

EVALUATE: To check the answers we confirm that the y -momentum, which was zero before the collision, is in fact zero after the collision. The y -momenta are

$$\begin{aligned}p_{A2y} &= (0.500 \text{ kg})(2.00 \text{ m/s})(\sin 36.9^\circ) = +0.600 \text{ kg} \cdot \text{m/s} \\ p_{B2y} &= -(0.300 \text{ kg})(4.47 \text{ m/s})(\sin 26.6^\circ) = -0.600 \text{ kg} \cdot \text{m/s}\end{aligned}$$

and their sum is indeed zero.

2.4.3 Collision- elastic and inelastic collision

Collisions in two dimension

Example 4.3.2:

The ballistic pendulum is a simple method to determine the speed. A bullet of mass m makes a completely inelastic collision with a block of wood of mass M , which is suspended like a pendulum. After the impact, the block swings up to a maximum height h . In terms of h , m , and M , what is the initial speed u of the bullet?



2.4.3 Collision- elastic and inelastic collision

Collisions in two dimension



Using conservation of momentum the speed **just** after collision can be calculated

$$mu = (m + M)v \quad \rightarrow \quad v = \frac{mu}{m + M}$$

As the pendulum swings upwards, the KE just after collision is converted into gravitational PE

$$\frac{1}{2}(m + M)v^2 = (m + M)gh$$

$$\frac{1}{2} \frac{m^2 u^2}{(m + M)^2} = gh$$

$$u = \sqrt{2gh} \left(1 + \frac{M}{m} \right)$$

2.4.4 Center of mass

For any collection of particles, we can calculate its total momentum. $p = m_1v_1 + m_2v_2$

$$= m_1 \frac{dx_1}{dt} + m_2 \frac{dx_2}{dt}.$$

We can pretend that this p is the momentum of an imaginary point particle of total mass

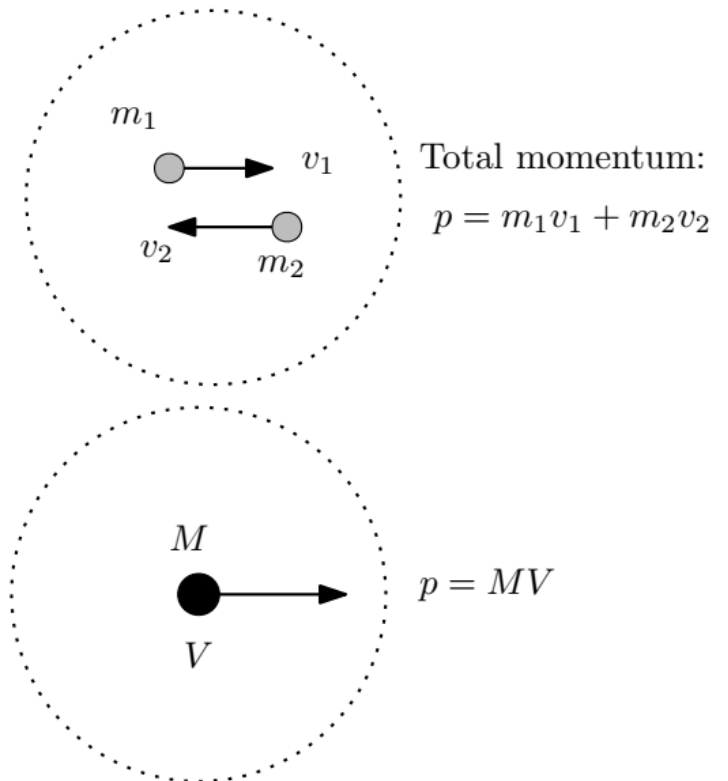
$$M = m_1 + m_2 \text{ with velocity } V: \quad p = MV$$
$$= (m_1 + m_2) \frac{dX}{dt}.$$

$$(m_1 + m_2) \frac{dX}{dt} = m_1 \frac{dx_1}{dt} + m_2 \frac{dx_2}{dt}$$

$$(m_1 + m_2)X = m_1x_1 + m_2x_2,$$

$$X = \frac{m_1x_1 + m_2x_2}{m_1 + m_2}$$

is called the **centre of mass**.



2.4.4 Center of mass

$$x_{\text{CM}} = \frac{1}{M} \sum_{i=1}^N m_i x_i, \quad M = \sum_{i=1}^N m_i = \text{total mass.}$$

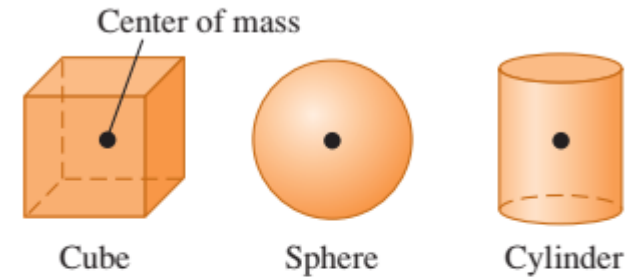
In 2D, the centre of mass coordinates are

$$x_{\text{CM}} = \frac{1}{M} \sum_{i=1}^N m_i x_i, \quad y_{\text{CM}} = \frac{1}{M} \sum_{i=1}^N m_i y_i$$

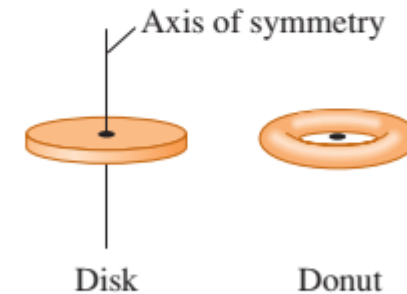
In vector form: $\vec{R}_{\text{CM}} = x_{\text{CM}} \hat{i} + y_{\text{CM}} \hat{j}$.

$$\vec{R}_{\text{CM}} = \frac{1}{M} \sum_{i=1}^N m_i \vec{r}_i.$$

8.28 Locating the center of mass of a symmetrical object.



If a homogeneous object has a geometric center, that is where the center of mass is located.



If an object has an axis of symmetry, the center of mass lies along it. As in the case of the donut, the center of mass may not be within the object.

2.4.4 Center of mass

Gravitational potential energy of a collection of particles: Consider a collection of many point particles with total mass M given by

$$M = m_1 + m_2 + \dots = \sum_i m_i.$$

The total gravitational potential energy is

$$\begin{aligned} U &= m_1gy_1 + m_2gy_2 + \dots = g(m_1y_1 + m_2y_2 + \dots) \\ &= Mg \frac{m_1y_1 + m_2y_2 + \dots}{M} \\ &= Mgy_{\text{CM}}. \end{aligned}$$

2.4.4 Center of mass

Centre of mass of extended bodies

We treat an extended body of mass M as a collection of tiny particles of mass dm .

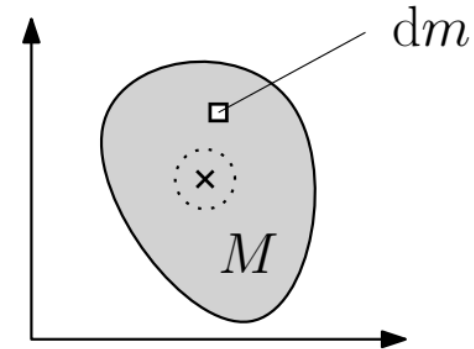
When summing up many infinitesimal quantities, summation becomes an integration (Riemann sum):

$$R_{\text{CM}} = \frac{1}{M} \sum_{i=1}^N m_i \vec{r}_i \rightarrow \frac{1}{M} \int dm \vec{r}$$

$$R_{\text{CM}} = \frac{1}{M} \int dm \vec{r},$$

where M is the total mass calculated by

$$M = \int dm.$$



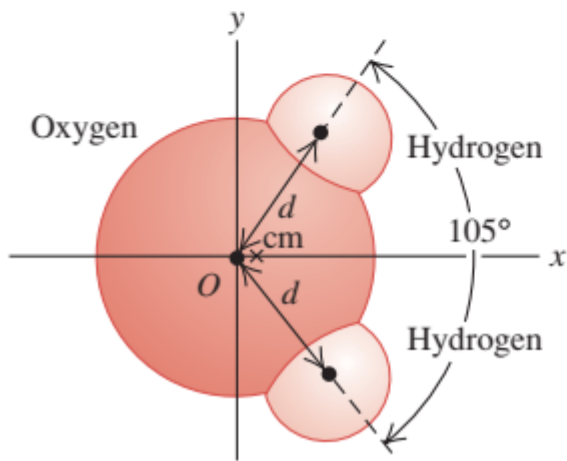
2.4.4 Center of mass

Figure 8.27 shows a simple model of a water molecule. The oxygen-hydrogen separation is $d = 9.57 \times 10^{-11}$ m. Each hydrogen atom has mass 1.0 u, and the oxygen atom has mass 16.0 u. Find the position of the center of mass.

SOLUTION

IDENTIFY and SET UP: Nearly all the mass of each atom is concentrated in its nucleus, whose radius is only about 10^{-5} times the overall radius of the atom. Hence we can safely represent each atom as a point particle. Figure 8.27 shows our coordinate system, with

8.27 Where is the center of mass of a water molecule?



the x -axis chosen to lie along the molecule's symmetry axis. We'll use Eqs. (8.28) to find x_{cm} and y_{cm} .

EXECUTE: The oxygen atom is at $x = 0$, $y = 0$. The x -coordinate of each hydrogen atom is $d \cos(105^\circ/2)$; the y -coordinates are $\pm d \sin(105^\circ/2)$. From Eqs. (8.28),

$$x_{cm} = \frac{\left[(1.0 \text{ u})(d \cos 52.5^\circ) + (1.0 \text{ u}) \right] \times (d \cos 52.5^\circ) + (16.0 \text{ u})(0)}{1.0 \text{ u} + 1.0 \text{ u} + 16.0 \text{ u}} = 0.068d$$

$$y_{cm} = \frac{\left[(1.0 \text{ u})(d \sin 52.5^\circ) + (1.0 \text{ u}) \right] \times (-d \sin 52.5^\circ) + (16.0 \text{ u})(0)}{1.0 \text{ u} + 1.0 \text{ u} + 16.0 \text{ u}} = 0$$

Substituting $d = 9.57 \times 10^{-11}$ m, we find

$$x_{cm} = (0.068)(9.57 \times 10^{-11} \text{ m}) = 6.5 \times 10^{-12} \text{ m}$$

EVALUATE: The center of mass is much closer to the oxygen atom (located at the origin) than to either hydrogen atom because the oxygen atom is much more massive. The center of mass lies along the molecule's *axis of symmetry*. If the molecule is rotated 180° around this axis, it looks exactly the same as before. The position of the center of mass can't be affected by this rotation, so it *must* lie on the axis of symmetry.

2.4.4 Center of mass

External Forces and Center-of-Mass Motion

$$M\vec{a}_{\text{cm}} = m_1\vec{a}_1 + m_2\vec{a}_2 + m_3\vec{a}_3 + \dots$$

$$\sum \vec{F} = \sum \vec{F}_{\text{ext}} + \sum \vec{F}_{\text{int}} = M\vec{a}_{\text{cm}}$$

Because of Newton's third law, the internal forces all cancel in pairs

$$\sum \vec{F}_{\text{ext}} = M\vec{a}_{\text{cm}} \quad (\text{body or collection of particles})$$

When a body or a collection of particles is acted on by external forces, the center of mass moves just as though all the mass were concentrated at that point and it were acted on by a net force equal to the sum of the external forces on the system.

2.4.4 Center of mass

External Forces and Center-of-Mass Motion

A rocket is propelled forward by rearward ejection of burned fuel that initially was in the rocket (which is why rocket fuel is also called *propellant*). The forward force on the rocket is the reaction to the backward force on the ejected material. The total mass of the system is constant, but the mass of the rocket itself decreases as material is ejected.

Application **Jet Propulsion in Squids**

Both a jet engine and a squid use variations in their mass to provide propulsion: They increase their mass by taking in fluid at low speed (air for a jet engine, water for a squid), then decrease their mass by ejecting that fluid at high speed. The net result is a propulsive force.



8.33 To provide enough thrust to lift its payload into space, this Atlas V launch vehicle ejects more than 1000 kg of burned fuel per second at speeds of nearly 4000 m/s.

